Probability & Statistics

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

What is the value of $16^{\frac{3}{4}}$?

By the definition of rational exponents, $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$ for any positive real number a. Here, $a = 16$, $m = 3$, and $n = 4$: $16^{\frac{3}{4}} = (\sqrt[4]{16})^3$ Since $2^4 = 16$, the principal fourth root is $\sqrt[4]{16} = 2$. Raising this result to the power of 3 gives: $2^3 = 8$ Common misconceptions: - Multiplying the base by the fraction gives $16 \cdot \frac{3}{4} = 12$, which incorrectly treats the exponent as a multiplier. - Finding $\sqrt[4]{16} = 2$ and then multiplying by 3.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Which expression is equivalent to $(16x^8y^4)^{\frac{3}{4}}$ for all positive values of x and y?

To simplify the expression $(16x^8y^4)^{\frac{3}{4}}$, apply the power of a product rule $(abc)^n = a^n b^n c^n$ and the power of a power rule $(a^m)^n = a^{mn}$: 1. Evaluate the numerical coefficient: $16^{\frac{3}{4}} = \left(\sqrt[4]{16}\right)^3 = 2^3 = 8$ 2. Apply the exponent to the variable x: $(x^8)^{\frac{3}{4}} = x^{8 \cdot \frac{3}{4}} = x^{\frac{24}{4}} = x^6$ 3. Apply the exponent to the variable y.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

What are the solutions to the quadratic equation $x^2 - 6x + 13 = 0$?

To solve the quadratic equation $x^2 - 6x + 13 = 0$, use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ with $a = 1$, $b = -6$, and $c = 13$: $x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(13)}}{2(1)} = \frac{6 \pm \sqrt{36 - 52}}{2} = \frac{6 \pm \sqrt{-16}}{2}$ Since $\sqrt{-16} = 4i$, simplify the expression: $x = \frac{6 \pm 4i}{2} = 3 \pm 2i$ - $x = 3 \pm 4i$ occurs if the radical term is not divided by the denominator 2. - $x = -3 \pm 2i$ occurs from a sign error when computing $-b$..

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

What are the center and radius of the circle given by the equation $(x - 2)^2 + (y + 6)^2 = 36$?

The standard form for the equation of a circle is: $(x - h)^2 + (y - k)^2 = r^2$ where (h, k) is the center of the circle and r is the radius. Comparing $(x - 2)^2 + (y + 6)^2 = 36$ to standard form: 1. $x - h = x - 2 \Rightarrow h = 2$ 2. $y - k = y - (-6) \Rightarrow k = -6$ 3. $r^2 = 36 \Rightarrow r = \sqrt{36} = 6$ Thus, the center is $(2, -6)$ and the radius is 6. Common errors: - Reversing the signs of the coordinates leads to $(-2, 6)$. - Forgetting to take the square root of the constant term leads t.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A sequence is defined by the recursive formula $a_n = 2a_{n-1} + 1$, with the first term $a_1 = 3$. What is the value of the third term, a_3?

The recursive formula for the sequence is $a_n = 2a_{n-1} + 1$, and the first term is $a_1 = 3$. To find the second term, a_2, we substitute $n = 2$ into the formula: $a_2 = 2a_{2-1} + 1 = 2a_1 + 1$ Since $a_1 = 3$, we have: $a_2 = 2(3) + 1 = 6 + 1 = 7$ To find the third term, a_3, we substitute $n = 3$ into the formula: $a_3 = 2a_{3-1} + 1 = 2a_2 + 1$ Since $a_2 = 7$, we have: $a_3 = 2(7) + 1 = 14 + 1 = 15$

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic equation $x^2 - 8x + 15 = 0$. What are the roots of this equation?

To find the roots of the quadratic equation $x^2 - 8x + 15 = 0$, we can factor the quadratic expression. We are looking for two numbers that multiply to 15 and add up to -8. These numbers are -3 and -5. Therefore, the factored form of the equation is $(x - 3)(x - 5) = 0$. Setting each factor to zero, we get $x - 3 = 0$ which gives $x = 3$, and $x - 5 = 0$ which gives $x = 5$. Thus, the roots are $x = 3$ and $x = 5$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the center of the circle represented by the equation $(x - 3)^2 + (y + 5)^2 = 16$?

The standard form of a circle's equation is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle. In the given equation, $(x - 3)^2 + (y + 5)^2 = 16$, we can identify $h = 3$ and $k = -5$. Therefore, the center of the circle is $(3, -5)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 3x + 4$ in describing shapes of data distributions, evaluate $f(5)$.

Final Answer: 19. Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for the normal distribution consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 16.5. The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A committee of 4 members is to be selected at random from a club of 7 seniors and 5 juniors. (a) What is the total number of possible committees? (b) Find the probability that the committee contains exactly 2 seniors and 2 juniors.

(a) Order does not matter, so the total number of ways to choose 4 members from 12 total people is: $\text{Total} = \binom{12}{4} = \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1} = 495.$ (b) Number of ways to choose 2 seniors from 7: $\binom{7}{2} = \frac{7 \times 6}{2 \times 1} = 21.$ Number of ways to choose 2 juniors from 5: $\binom{5}{2} = \frac{5 \times 4}{2 \times 1} = 10.$ By the fundamental counting principle, favorable outcomes: $21 \times 10 = 210.$ Probability.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Probability & Statistics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In a class of 200 students taking a test with mean 65 and SD 12 (normally distributed), how many students would you expect to score above 80?

$P(X > 80) = P(Z > 1.25) = 1 - 0.8944 = 0.1056$. Expected: $200 \times 0.1056 \approx 21$ students.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Solve the rational equation for x: $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$. Check for extraneous solutions.

Factor the quadratic denominator: $x^2 + x - 6 = (x-2)(x+3)$. The LCD is $(x-2)(x+3)$, with restricted domain $x \neq 2, -3$. Multiply both sides by $(x-2)(x+3)$: $x(x+3) + 1(x-2) = 5$.
$x^2 + 3x + x - 2 = 5 \Rightarrow x^2 + 4x - 7 = 0.$ Apply the quadratic formula:
$x = \frac{-4 \pm \sqrt{16 - 4(1)(-7)}}{2} = \frac{-4 \pm \sqrt{16+28}}{2} = \frac{-4 \pm \sqrt{44}}{2} = \frac{-4 \pm 2\sqrt{11}}{2} = -2 \pm \sqrt{11}.$ Check restrictions: $-2 \pm \sqrt{11} \approx 1.317$ and -5.317. Neither equals.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 6.01 Radicals & Rational Exponents**

Solve the linear system using matrix inversion or Cramer's Rule: $3x - 2y = 8$ $5x + 4y = 6$

Write the system in matrix form $A\mathbf{x} = \mathbf{b}$: $(3, -2; 5, 4)(x; y) = (8; 6).$ Calculate the determinant of coefficient matrix A: $\det A = (3)(4) - (-2)(5) = 12 - (-10) = 22.$ Find the inverse matrix A^{-1}: $A^{-1} = \frac{1}{22}(4, 2; -5, 3).$ Multiply $A^{-1}\mathbf{b}$.

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16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 7.01 Sequences & Series**

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.

Write $y = \frac{3x-1}{2x+5}$. The domain of f is $x \neq -2.5$. Its horizontal asymptote is $y = 3/2$, so the range of f is $y \neq 1.5$. Swap x and y and solve for y: $x = \frac{3y-1}{2y+5} \Rightarrow x(2y+5) = 3y-1$.
$2xy + 5x = 3y - 1 \Rightarrow 2xy - 3y = -5x - 1$. $y(2x-3) = -(5x+1) \Rightarrow y = \frac{-5x-1}{2x-3} = \frac{5x+1}{3-2x}$. Inverse function: $f^{-1}(x) = \frac{5x+1}{3-2x}$. Domain of f^{-1}: $x \neq \frac{3}{2}$ (the range of f). Range of f^{-1}.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 8.01 Conic Sections**

If $\sin\theta = -\frac{4}{5}$ and θ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Using the Pythagorean identity $\sin^2\theta + \cos^2\theta = 1$: $\left(-\frac{4}{5}\right)^2 + \cos^2\theta = 1 \Rightarrow \frac{16}{25} + \cos^2\theta = 1 \Rightarrow \cos^2\theta = \frac{9}{25}$.
Since θ is in Quadrant III, cosine is negative: $\cos\theta = -\frac{3}{5}$. Tangent is positive in Quadrant III: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-4/5}{-3/5} = \frac{4}{3}$. Double-angle formula for sine.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 9.01 Trigonometric Functions**

Solve $2x^2 - 6x + 5 = 0$ by completing the square, expressing all solutions in exact $a \pm bi$ form.

Divide by the leading coefficient 2 or factor it out from the variable terms:
$2(x^2 - 3x) = -5 \Rightarrow x^2 - 3x = -\frac{5}{2}$. Add $\left(-\frac{3}{2}\right)^2 = \frac{9}{4}$ to both sides:
$x^2 - 3x + \frac{9}{4} = -\frac{5}{2} + \frac{9}{4} = -\frac{10}{4} + \frac{9}{4} = -\frac{1}{4}$. Write as a perfect square and take the square root of both sides:
$\left(x - \frac{3}{2}\right)^2 = -\frac{1}{4} \Rightarrow x - \frac{3}{2} = \pm\sqrt{-\frac{1}{4}} = \pm\frac{i}{2}$. $x = \frac{3}{2} \pm \frac{1}{2}i$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 1.01 Functions & Their Graphs**

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

By the Rational Root Theorem, possible rational roots are $\pm\{1, 3\}/\{1, 2\} = \pm\{1, 3, 1/2, 3/2\}$. Test $x = 1$: $P(1) = 2(1)^3 + 3(1)^2 - 8(1) + 3 = 2 + 3 - 8 + 3 = 0$. ✓ So $(x-1)$ is a factor. Divide $P(x)$ by $(x-1)$ using synthetic division: $2x^3 + 3x^2 - 8x + 3 = (x-1)(2x^2 + 5x - 3)$. Factor the remaining quadratic:
$2x^2 + 5x - 3 = (2x-1)(x+3)$. Setting each factor to zero yields the complete set of roots:
$x = 1, \quad x = \frac{1}{2}, \quad x = -3$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 10.01 Matrices & Systems**

Solve the radical equation $\sqrt{2x+7} - x = 2$ for all real solutions and confirm extraneous roots.

Isolate the radical term: $\sqrt{2x+7} = x + 2$. Square both sides: $2x + 7 = (x+2)^2 = x^2 + 4x + 4$.
$x^2 + 2x - 3 = 0 \Rightarrow (x+3)(x-1) = 0 \Rightarrow x = -3$ or $x = 1$. Test in the original un-squared equation:Check $x = -3$:
$\sqrt{2(-3)+7} - (-3) = \sqrt{1} + 3 = 1 + 3 = 4 \neq 2$. (Extraneous ✗) Check $x = 1$:
$\sqrt{2(1)+7} - 1 = \sqrt{9} - 1 = 3 - 1 = 2$. (Valid ✓) The only valid real solution is $x = 1$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(2)$.

a. 21 ✓

b. 27

c. 12

d. 25

Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(2)$.

a. 7 ✓

b. 9

c. 4

d. 11

Substitute $x = 2$: $f(2) = 2(2) + 3 = 7$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for random sampling consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for estimating proportions from samples consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for describing shapes of data distributions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for interpreting and comparing z-scores consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in interpreting and comparing z-scores, evaluate $f(3)$.

a. 21 ✓

b. 26

c. 15

d. 25

Substitute $x = 3$: $f(3) = 5(3) + 6 = 21$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for normal distribution probabilities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in normal distribution probabilities, evaluate $f(3)$.

a. 25 ✓

b. 31

c. 18

d. 29

Substitute $x = 3$: $f(3) = 6(3) + 7 = 25$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A fair six-sided die is rolled twice. Let X be the random variable representing the sum of the outcomes of the two rolls. The probability distribution of X is given in the table below, where p is an unknown probability. $Sum\ (x), 2, 3, 4, 5, 6, 7; P(X = x), 136, p, 336, 436, 536, 636;$ What is the value of p?

a. $\frac{2}{36}$ ✓

b. $\frac{5}{36}$

c. $\frac{4}{36}$

d. $\frac{3}{36}$

The sum of all probabilities in a probability distribution must equal 1. We are given the probabilities for several sums, and we need to find the unknown probability p for the sum of 3. The sum of the given probabilities is: $P(X = 2) + P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) = \frac{1}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} = \frac{1+3+4+5+6}{36} = \frac{19}{36}$ Since the total probability must be 1, we have: $P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) = 1$ Substituting the known value.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The weights of apples harvested in an orchard are approximately normally distributed with mean μ and standard deviation σ. According to the empirical rule (68-95-99.7% rule), approximately 84% of the apples have a weight greater than 166 grams, and approximately 2.5% of the apples have a weight greater than 214 grams. What is the mean weight μ, in grams, of the apples?

a. 178 grams

b. **182 grams** ✓

c. 190 grams

d. 198 grams

According to the empirical rule (68-95-99.7% rule) for a normal distribution: 1. Approximately 68% of the data lies within 1 standard deviation of the mean ($\mu \pm \sigma$). Since the normal distribution is symmetric, 34% lies between $\mu - \sigma$ and μ. Adding the 50% that lies above μ, the proportion of data greater than $\mu - \sigma$ is $34\% + 50\% = 84\%$. Thus: $\mu - \sigma = 166$ 2. Approximately 95% of the data lies within 2 standard deviations of the mean ($\mu \pm 2\sigma$). The remaining 5%.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In a championship playoff, two teams, Team A and Team B, play a best-of-five series, where the first team to win three games wins the series. The games are independent, cannot end in a tie, and in each individual game, the probability that Team A wins is p, where $0 < p < 1$ and $p \neq \frac{1}{2}$. Suppose the probability that the series is decided in exactly 3 games is equal to the probability that the series requires a decisive 5th game. What is the value of the product $p(1-p)$?

a. $\frac{\sqrt{33}-3}{12}$ ✓

b. $\frac{\sqrt{33}+3}{12}$

c. $\frac{\sqrt{33}-3}{6}$

d. $\frac{6-\sqrt{33}}{12}$

Let $q = 1 - p$ be the probability that Team B wins any individual game. 1. **Probability that the series ends in exactly 3 games:** This occurs if Team A sweeps $3-0$ or Team B sweeps $3-0$: $P(\text{3 games}) = p^3 + q^3$ Using the algebraic sum of cubes identity: $p^3 + q^3 = (p+q)(p^2 - pq + q^2) = (p+q)((p+q)^2 - 3pq)$ Since $p + q = 1$, setting $x = pq = p(1-p)$ gives: $P(\text{3 games}) = 1 - 3pq = 1 - 3x$ 2. **Probability that the series requires a 5th game:** A 5th game is played if and only if.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A set of test scores is normally distributed with a mean of $\mu = 500$ and a standard deviation of $\sigma = 100$. According to the empirical rule (the $68-95-99.7\%$ rule), approximately what percentage of scores fall between 400 and 700?

a. 81.5% ✓

b. 68.0%

c. 84.0%

d. 95.0%

According to the empirical rule for a normal distribution: 1. Approximately 68% of the data lies within 1 standard deviation of the mean ($\mu \pm 1\sigma$). Because the distribution is symmetric, 34% lies between $\mu - 1\sigma$ and μ. 2. Approximately 95% of the data lies within 2 standard deviations of the mean ($\mu \pm 2\sigma$). By symmetry, 47.5% lies between μ and $\mu + 2\sigma$. Identify the given boundaries relative to $\mu = 500$ and $\sigma = 100$: - Lower bound.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A factory produces two types of components, A and B, using two machines, M1 and M2. The time (in minutes) required to produce one unit of each component on each machine is given in the table below: | Component | Machine M1 | Machine M2 | |---|---|---| | A | 2 | 3 | | B | 4 | 1 | The factory has 480 minutes of M1 time and 240 minutes of M2 time available per day. Let x be the number of units of component A produced per day and y be the number of units of component B produced per day. Suppose the profit from selling one unit of component A is 5 and the profit from selling one unit of component B is 8. The factory wants to maximize its total daily profit. Formulate this problem as a linear programming problem, defining the objective function and the constraints.

a. **Maximize $P = 5x + 8y$ subject to $2x + 4y \le 480$, $3x + y \le 240$, $x \ge 0$, $y \ge 0$.** ✓

b. Maximize $P = 8x + 5y$ subject to $2x + 3y \le 480$, $4x + y \le 240$, $x \ge 0$, $y \ge 0$.

c. Maximize $P = 5x + 8y$ subject to $4x + 2y \le 480$, $y + 3x \le 240$, $x \ge 0$, $y \ge 0$.

d. Maximize $P = 5x + 8y$ subject to $2x + 4y \le 240$, $3x + y \le 480$, $x \ge 0$, $y \ge 0$.

The objective is to maximize the total daily profit. The profit from selling x units of component A is $5x$ and the profit from selling y units of component B is $8y$. Therefore, the objective function is $P = 5x + 8y$. The constraints are based on the available machine time. For Machine M1, producing x units of A takes $2x$ minutes and producing y units of B takes $4y$ minutes. The total time on M1 must be less than or equal to the available 480 minutes: $2x + 4y \le 480$. For Machine M2, producing x un.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A bag contains 5 red marbles and 3 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

a. $\frac{3}{8}$

b. $\frac{5}{8}$ ✓

c. $\frac{1}{2}$

d. $\frac{5}{3}$

The total number of marbles in the bag is the sum of red marbles and blue marbles: $5 + 3 = 8$. The number of favorable outcomes (drawing a red marble) is 5. The probability of an event is calculated as the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of drawing a red marble is $\frac{\text{Number of red marbles}}{\text{Total number of marbles}} = \frac{5}{8}$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Prove the weak law of large numbers: $\bar{X}_n \to \mu$ as $n \to \infty$.

a. **By Chebyshev's inequality:** $P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0$. ✓✓

b. Not provable

c. Use the CLT only

d. Only for normal distributions

Applying Probability & Statistics principles, Standard proof via Chebyshev. Therefore, the verified correct option is By Chebyshev's inequality: $P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0$. ✓.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The moment generating function uniquely determines the distribution. Why?

a. Only if $E(X)$ is finite

b. Not always true

c. Only for discrete distributions

d. **Because the MGF encodes all moments and there is a one-to-one correspondence between MGFs and distributions (under regularity conditions)** ✓

Applying Probability & Statistics principles, Uniqueness theorem for MGFs. Therefore, the verified correct option is Because the MGF encodes all moments and there is a one-to-one correspondence between MGFs and distributions (under regularity conditions).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Trigonometric Functions

State and prove the Bayes' theorem for continuous distributions.

a. Not valid for continuous distributions

b. Same as discrete case

c. $f(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)d\theta}$ **— from the definition of conditional density** ✓

d. Use the CLT

Applying Probability & Statistics principles, Continuous Bayes' theorem. Therefore, the verified correct option is $f(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)d\theta}$ — from the definition of conditional density.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Functions & Their Graphs

Show that the binomial distribution approaches Poisson as $n \to \infty$, $p \to 0$, $np = \lambda$ fixed.

a. Requires $n > 100$

b. Not true

c. $\binom{n}{k}p^k(1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ **— use** $(1-\lambda/n)^n \to e^{-\lambda}$ ✓

d. Only for $k = 0$

Applying Probability & Statistics principles, Poisson limit theorem. Therefore, the verified correct option is $\binom{n}{k}p^k(1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ — use $(1-\lambda/n)^n \to e^{-\lambda}$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Matrices & Systems

State the definition of a sufficient statistic.

a. $T(X)$ **is sufficient for** θ **if the conditional distribution of** X **given** $T(X)$ **does not depend on** θ ✓

b. It must be unbiased

c. $T(X) = \theta$ always

d. Only the sample mean is sufficient

Applying Probability & Statistics principles, Sufficiency definition. Therefore, the verified correct option is $T(X)$ is sufficient for θ if the conditional distribution of X given $T(X)$ does not depend on θ.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

State and apply the Central Limit Theorem.

a. Only for discrete distributions

b. **For large n, $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately, regardless of the population distribution** ✓

c. Only for $n > 1000$

d. Only for normal populations

Applying Probability & Statistics principles, CLT statement. Therefore, the verified correct option is For large n, $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately, regardless of the population distribution.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Polynomial Functions

In a hypothesis test, the p-value is:

a. The probability that H_0 is true

b. **The probability of observing a test statistic as extreme as observed, assuming H_0** ✓

c. The type II error rate

d. The significance level

Applying Probability & Statistics principles, Definition of p-value. Therefore, the verified correct option is The probability of observing a test statistic as extreme as observed, assuming H_0.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Rational Expressions

State the Fisher information: $I(\theta) =$

a. $E\left[\frac{\partial^2}{\partial\theta^2}\ln f\right]$

b. $E\left[\left(\frac{\partial}{\partial\theta}\ln f(X;\theta)\right)^2\right]$ ✓

c. $\frac{1}{\mathrm{Var}(\hat{\theta})}$

d. $\mathrm{Var}(\hat{\theta})$

Applying Probability & Statistics principles, Fisher information definition. Therefore, the verified correct option is $E\left[\left(\frac{\partial}{\partial\theta}\ln f(X;\theta)\right)^2\right]$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Show the mgf of $\bar{X}$ for i.i.d. X_i with mgf $M(t)$ is $[M(t/n)]^n$.

a. $M_{\bar{X}}(t) = E[e^{t\bar{X}}] = E\left[\prod e^{tX_i/n}\right] = \prod M(t/n) = [M(t/n)]^n$ ✓

b. Only for normal X

c. Only for $n = 2$

d. Not true

Applying Probability & Statistics principles, MGF of sample mean. Therefore, the verified correct option is $M_{\bar{X}}(t) = E[e^{t\bar{X}}] = E\left[\prod e^{tX_i/n}\right] = \prod M(t/n) = [M(t/n)]^n$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Radicals & Rational Exponents

For two random variables X, Y, prove $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$.

a. Only if independent

b. Only for discrete variables

c. $E[(X+Y-\mu_X-\mu_Y)^2] = E[(X-\mu_X)^2] + 2E[(X-\mu_X)(Y-\mu_Y)] + E[(Y-\mu_Y)^2]$. ✓ ✓

d. Not true

Applying Probability & Statistics principles, Variance of sum formula. Therefore, the verified correct option is $E[(X+Y-\mu_X-\mu_Y)^2] = E[(X-\mu_X)^2] + 2E[(X-\mu_X)(Y-\mu_Y)] + E[(Y-\mu_Y)^2]$. ✓.

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